Automated Network Request Management

Phase 2: Project Design Phase — System Architecture, Data Schema, UI Policies & Relationships

1. System Architecture & Data Model Overview

The Technical Architecture of the Automated Network Request Management solution combines standard platform catalog items with a custom, dedicated database table **Database Tables (u_database_tables)**. This separation ensures operational clarity, strict data governance, and high performance without polluting core platform tables.

> **Architectural Flow:** End-User Catalog Submission (Service Portal) ➔ Requested Item (sc_req_item) ➔ Flow Designer Orchestration ➔ Custom Transaction Table (u_database_tables) ➔ Approval Request Relationship (sysapproval_approver) ➔ Automated Task & Email Dispatch.

2. Custom Database Table Specification (u_database_tables)

A custom table labeled **Database Tables** with system name u_database_tables was defined in the Global scope. The schema captures critical parameters of each network engagement including financial amounts, connection specifics, contact info, and approval states.

Field Label	Field Name	Data Type	Length	Default / Logic
Number	u_number	String	40	javascript:global.getNextObjNumberPadded();
Database Number	u_database_number	String	32	Manual / System identifier
Requested For	u_requested_for	Reference (sys_user)	32	Dynamic from session requester
Assignment to	u_assignment_to	Reference (sys_user)	32	Fulfillment engineer / ITIL user
Mobile Number	u_mobile_number	String	40	Contact mobile validation
Type of Connection	u_type_of_connection	Choice / String	40	NEW / EXISTING
Total Amount	u_total_amount	String / Decimal	32	Financial impact value
Mode of Payment	u_mode_of_payment	Choice / String	40	UPI / Online / Credit
Customer Address	u_customer_address	String	40	Physical site / delivery location
Approval Status	u_approval_status	Choice / String	40	approved / waiting for approval / rejected

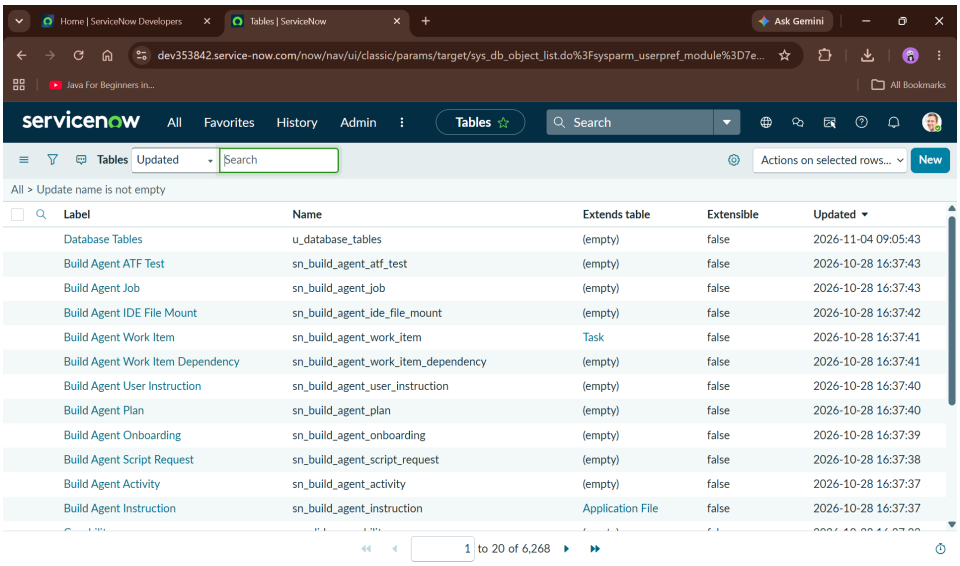


Figure 1: ServiceNow System Definition > Tables List showing u_database_tables registered in instance dev353842

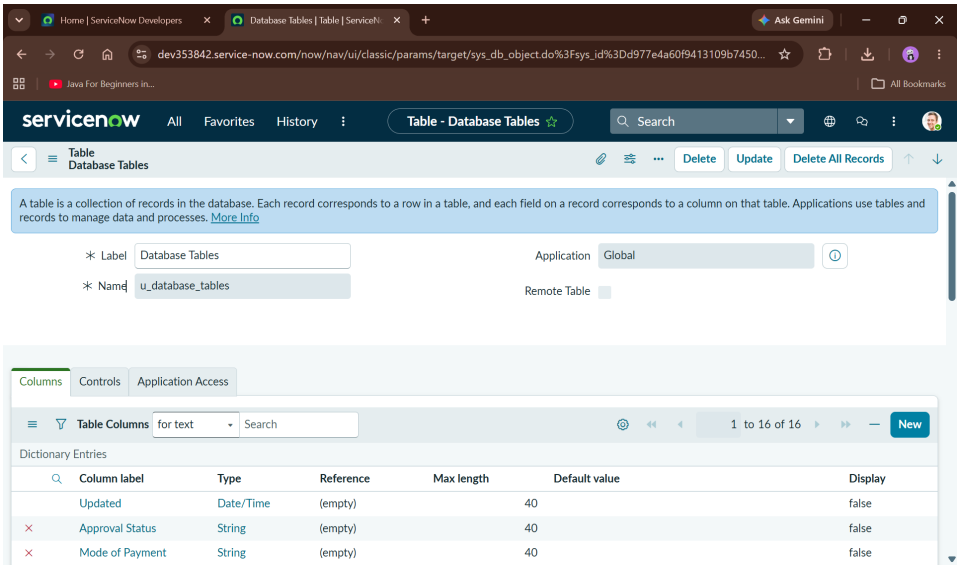


Figure 2: Table Configuration Form & Attributes for u_database_tables in Global Scope

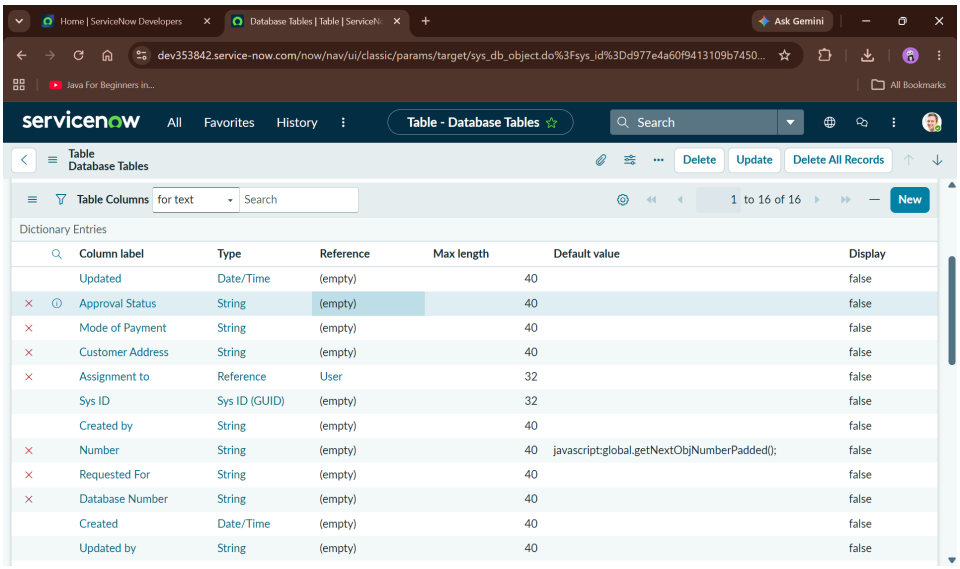


Figure 3: Dictionary Entries & Column Definitions for u_database_tables including Auto-Numbering

3. Form Design & Presentation Layout

The Form Designer was utilized to structure a clean, two-column form view displaying operational headers, requester details, and connection parameters. Mandatory constraints guarantee that requests cannot be saved without mandatory fields completed.

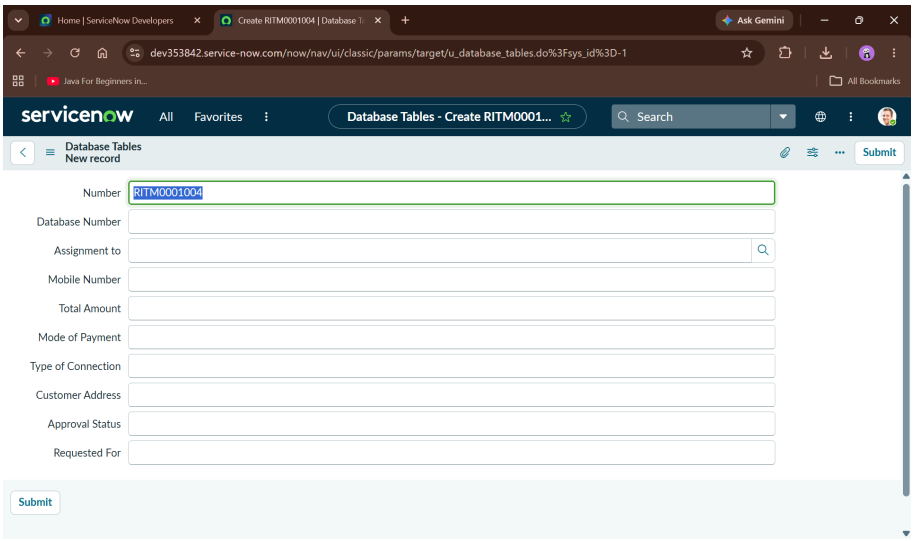


Figure 4: Form Designer Layout for Network Database Table (New Record State)

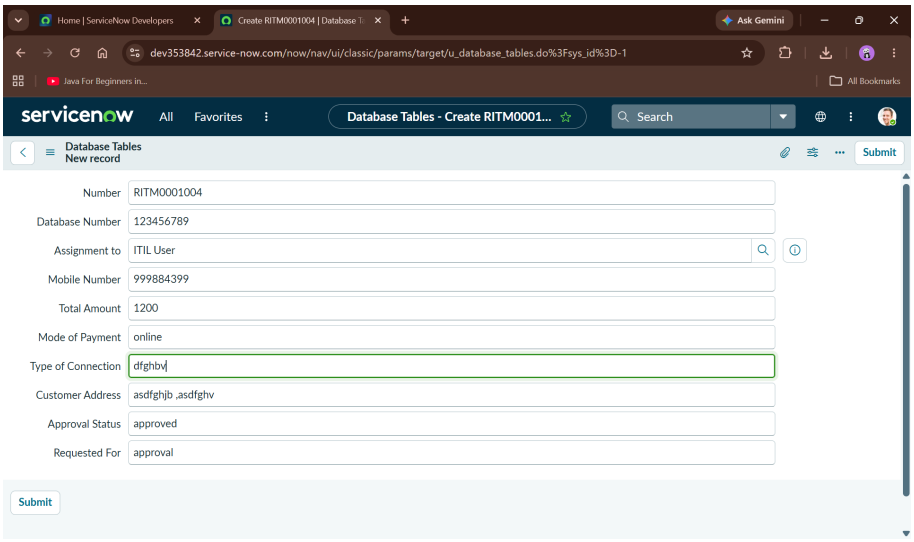


Figure 5: Form View with Populated Test Network Record (RITM0001004 / Number 123456789)

4. Approval Request Relationship & Related Lists Architecture

To enable approvers and fulfillers to see approvals directly inside the custom u_database_tables record, a platform Relationship was created linking u_database_tables to sysapproval_approver.

- **Navigate:** System Definition > Relationships > Click *New*.
- **Name:** Approval Request.
- **Applies to Table:** Database Tables [u_database_tables].
- **Queries from Table:** Approvals [sysapproval_approver].
- **Query Script:** current.addQuery('sysapproval', parent.sys_id);
- **Form Integration:** Add the Related List 'Approval Request' to the Database Tables form layout.

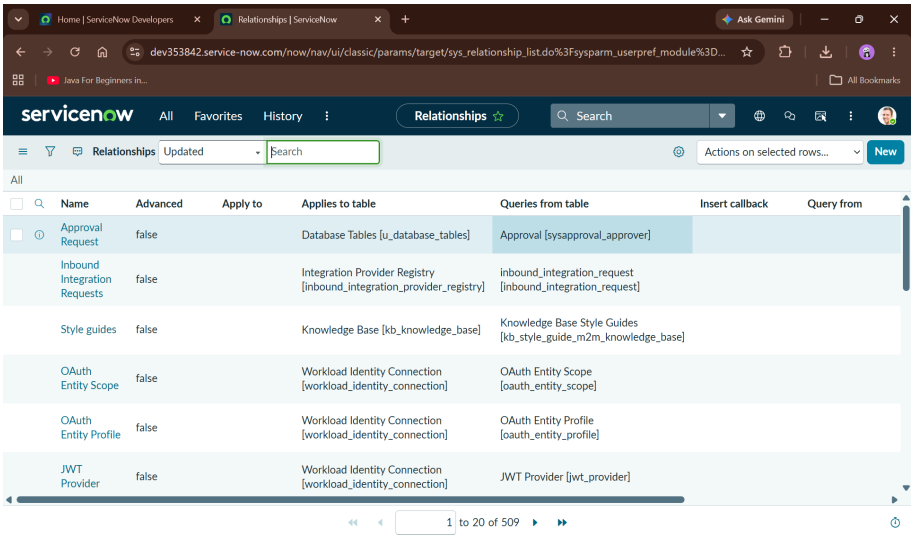


Figure 6: System Definition > Relationships List displaying registered Approval Request entity

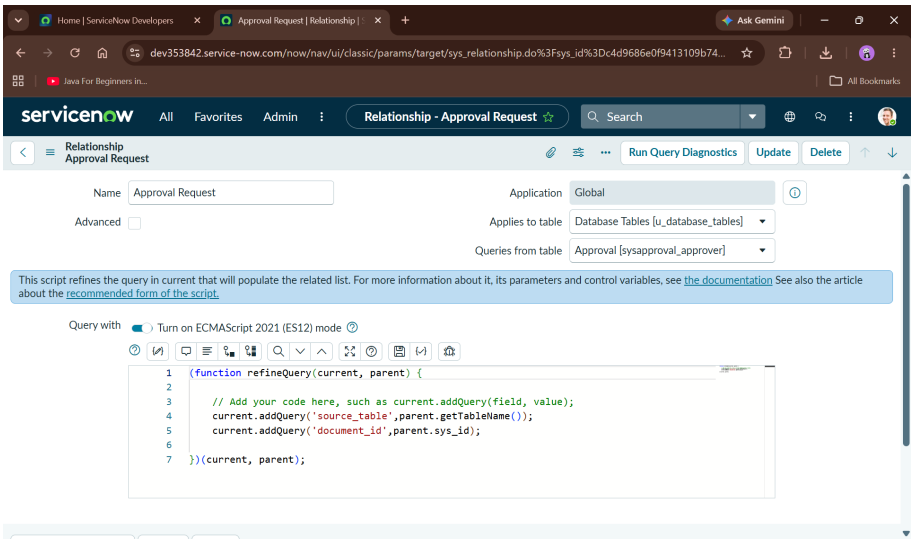


Figure 7: Approval Request Relationship Form Configuration

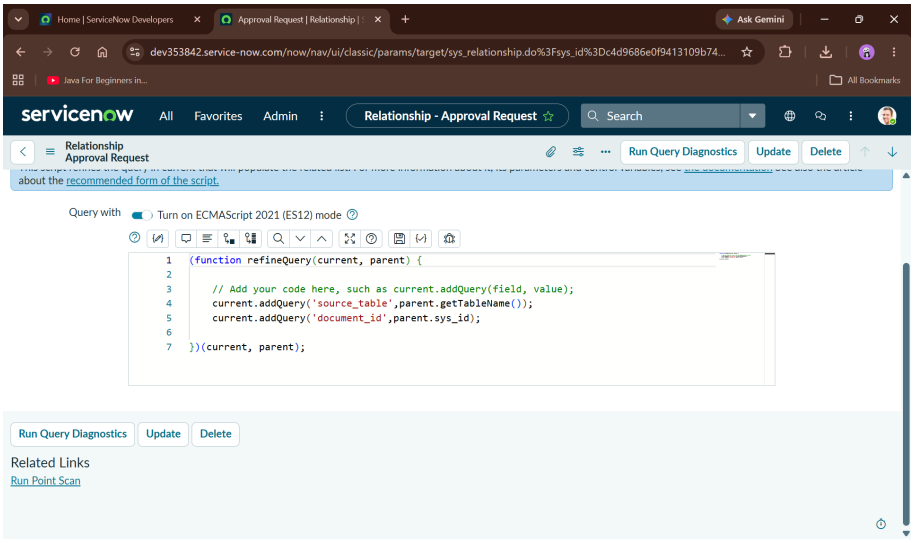


Figure 8: Query Script Configuration Linking sysapproval_approver to Parent u_database_tables Record

5. Service Catalog Item Design: 'Network Request'

A dedicated Catalog Item was created in the Service Catalog under category *Standard Changes / Network Services*. The catalog item encapsulates end-user ordering parameters, workflow bindings, and catalog portal configurations.

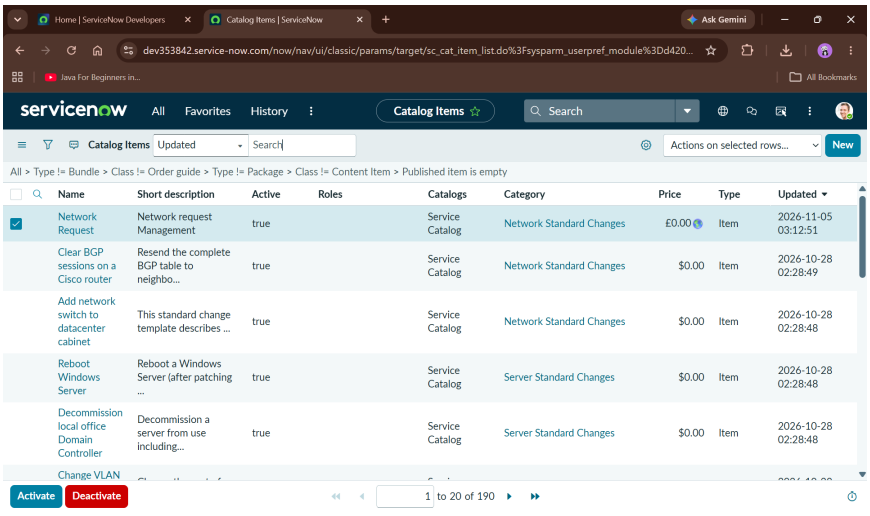


Figure 9: Service Catalog > Catalog Items Inventory Search

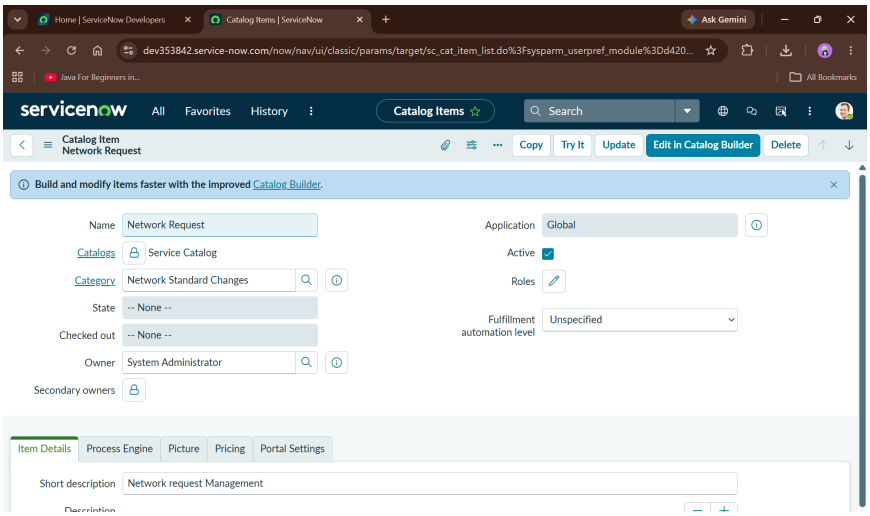


Figure 10: Catalog Items Registry displaying Active 'Network Request' Item

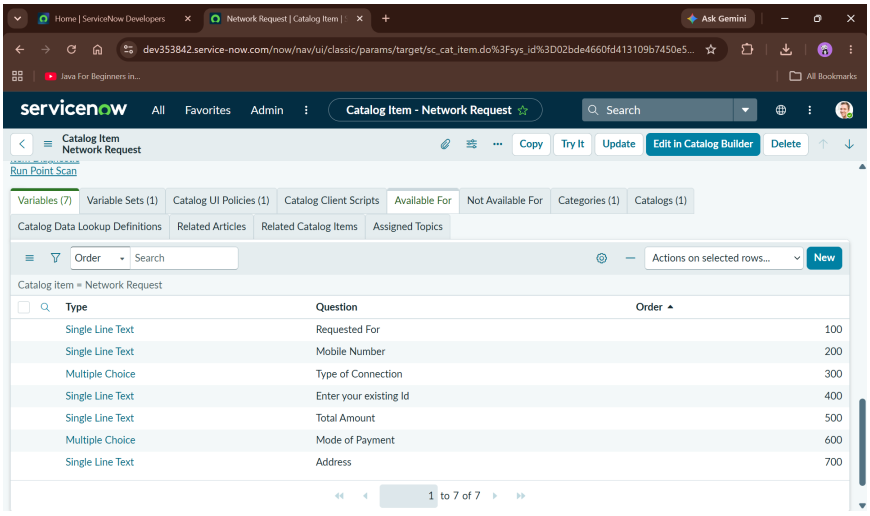


Figure 11: Network Request Catalog Item Configuration Form & Categorization

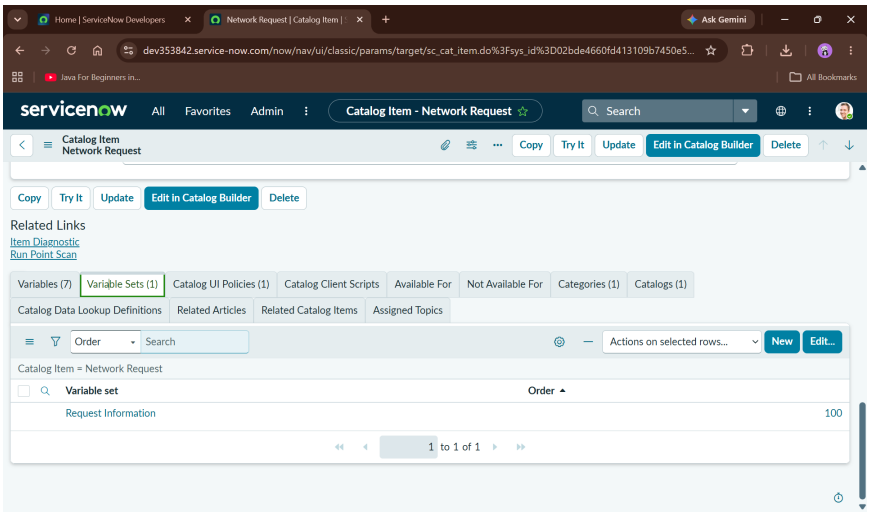


Figure 12: Catalog Item Related Lists — Process Engine Binding to 'Network Requests' Flow

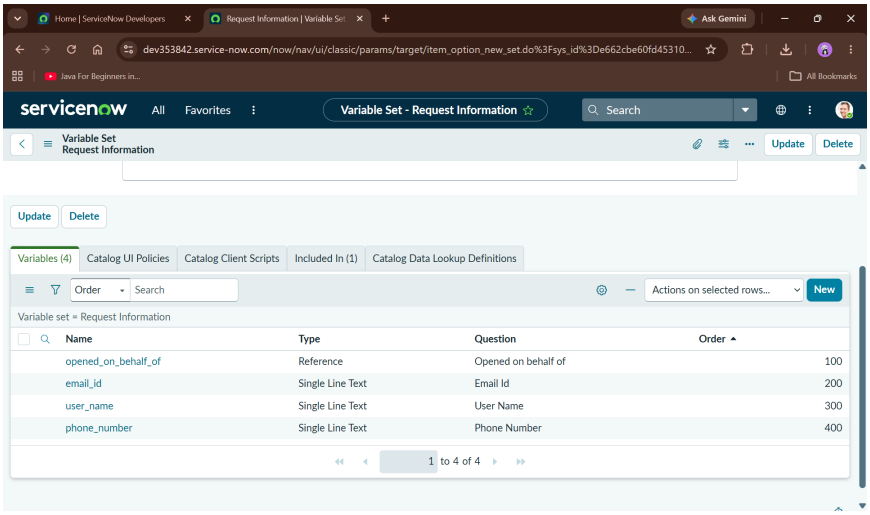


Figure 13: Variable Set Definition ('Request Information') containing Core Variable Data Model

6. Dynamic Catalog UI Policies & Conditional Visibility Logic

To streamline the user experience, Catalog UI Policies were engineered to display conditional variables dynamically without custom JavaScript.

> **Conditional Rule:** When Type of Connection is selected as **EXISTING**, dynamically display the field **Existing ID** and enforce validation. If NEW is selected, automatically hide the field and clear its value.

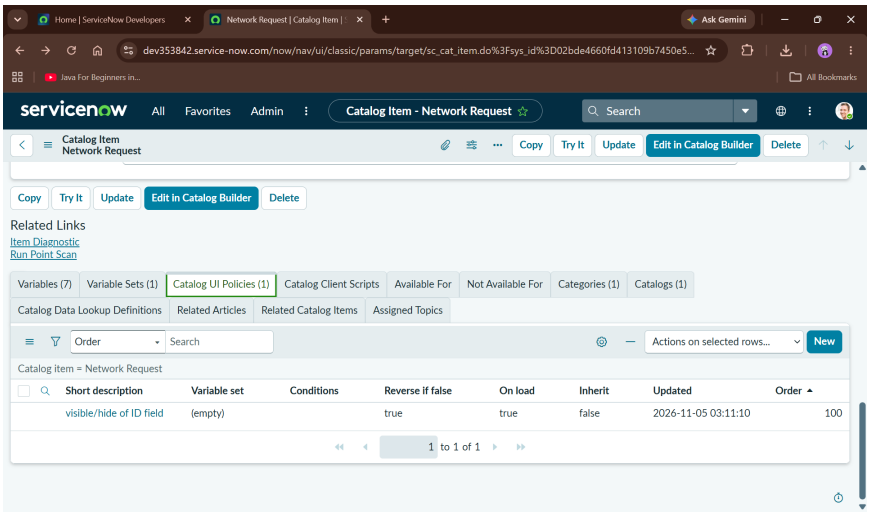


Figure 14: Catalog UI Policies Related List for 'Network Request' Item

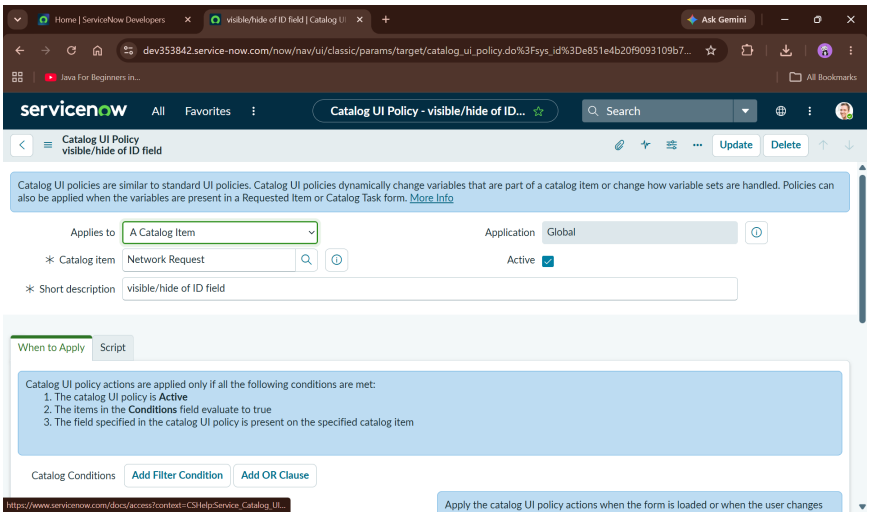


Figure 15: Catalog UI Policy 'visible/hide of ID field' Condition Definition (Connection = EXISTING)

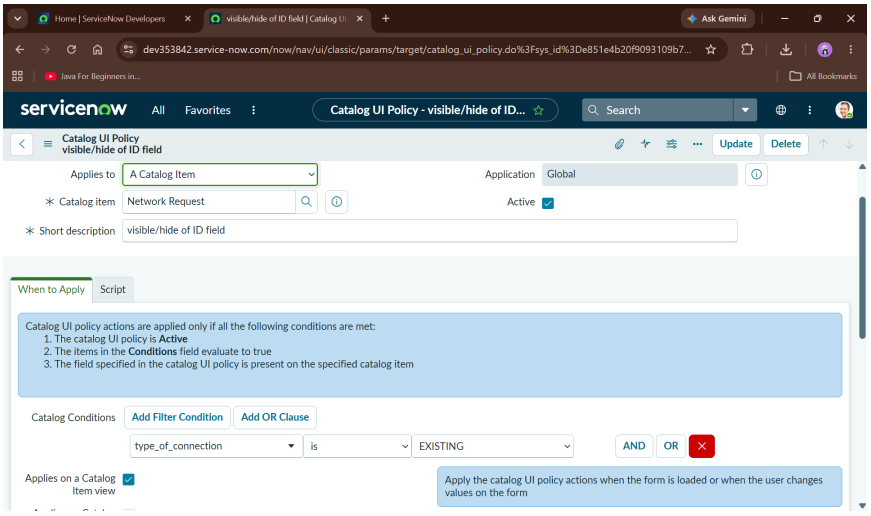


Figure 16: Catalog UI Policy Action: Existing ID Target Variable Set to Visible = True

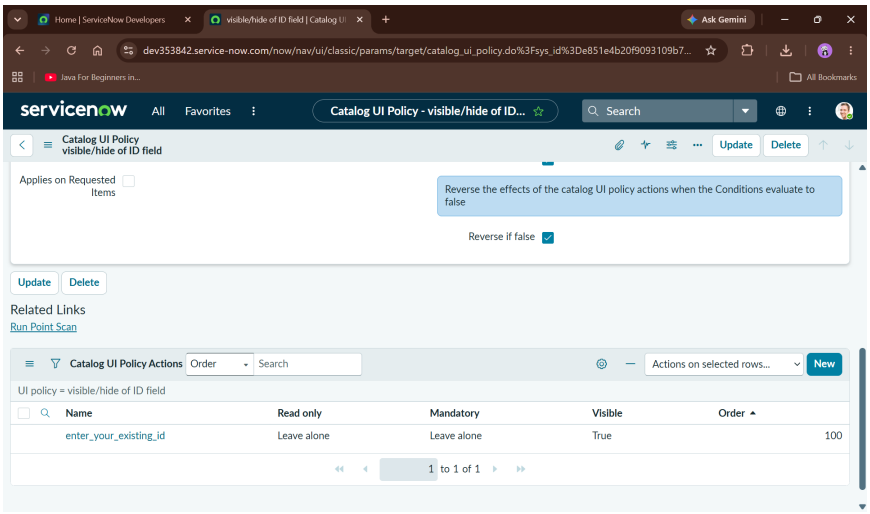


Figure 17: UI Policy Execution Rules (Applies on Load = True, Reverse if False = True)

Phase 2 Summary: The data structures, relationships, catalog items, variable sets, and dynamic UI policies are fully designed and validated. All artifacts are prepared for Phase 3 Project Planning and Phase 4 Flow Designer automation.